

Fig. 2: Actual-size PCB of the mixer circuit

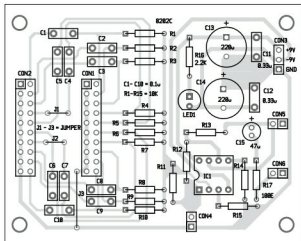


Fig. 3: Component layout of the PCB

circuit provides the inverted and the non-inverted sum of all input signals. Both outputs can be used simultaneously or individually. The outputs can drive loads as low as 400 ohms.

The DC component of the sum produced by second amplifier A2 is passed through DC-cutting capacitor C15 and is available on connector CON6.

The full power supply range of RC4560/NE5532 is available. This IC can work in the range of $\pm 4V$ to $\pm 18V$ (preferably up to $\pm 15V$). The quiescent current without signal is typically less than 10mA.

We have tested the circuit on $\pm 9V$ power supply. The circuit can also be powered using dry batteries, two 6V or 12V rechargeable accumulators or well-regulated mains power supply.

ately after its assembly, if done properly.

Construction and testing

An actual-size, single-side PCB for the mixer circuit is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box such that connectors CON1 and CON2 can be used to connect ten inputs. Ensure proper wiring to avoid any mistake.

Panel-mount the input and output interface, as per requirement. ●

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IC RC4560 can work with loads as low as 400 ohms and can drive audio lines directly. With IC NE5532 the loads should be 600 ohms or more. The power supply of NE5532 can go up to $\pm 20V$ and if the load is 2-kilo-ohm or more, the output voltage swing will be larger than with RC4560.

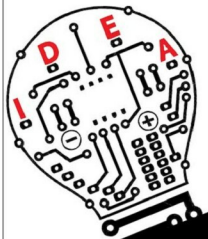
Usage of IC OPA2134 is also possible and will produce excellent results. If TL072 or TL082 are used in the circuit, output loads should be 2-kilo-ohm or more. This circuit of a simple AC and DC mixer, follower and buffer with ten inputs can work with signals of 0Hz to more than 100kHz, with a large variety of operational amplifiers.

The mixer does not contain any expensive or rare components and will work immedi-

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